ML INTERVIEW PREP

15

MACHINE LEARNING

interview questions you **MUST know**

The fundamentals asked again and again in real interviews. Most candidates fail on the basics — these make you sound confident, practical and senior.

Overfitting | Bias–Variance | ROC / AUC | Random Forest | Gradient Descent

Ashwini
Senior AI/ML Engineer

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FIRST, THE BASICS

What is Machine Learning?

A subset of Artificial Intelligence that teaches machines to **learn from data** and identify patterns to make predictions — without being manually programmed.

EXAMPLE — SPAM DETECTION

- We don't write rules to detect spam
- We show the system thousands of spam and non-spam emails
- It learns what spam looks like from the data

Most interviewers want clear thinking and practical understanding — not perfect math.

Ashwini
Senior AI/ML Engineer

What are the main types of Machine Learning?

Supervised Learning

Dataset has correct answers (labels). The model learns input → output. E.g. spam detection, price prediction.

Unsupervised Learning

Works with unlabeled data; finds hidden patterns on its own. E.g. clustering, grouping documents.

Semi-Supervised Learning

A small portion labeled, most unlabeled. Useful when labeling is costly (medical, legal).

Reinforcement Learning

An agent learns from an environment via rewards and penalties. E.g. game AI, robotics.

Interview tip: most real-world business problems start as supervised learning.

Difference between Classification and Regression?

The difference lies in **what the model predicts**.

Classification

Predicts a category or class — the output belongs to a fixed set of labels. E.g. spam or not, whether a customer will churn.

Regression

Predicts a continuous numerical value — the output can be any real number. E.g. house prices, temperature, future sales.

QUICK WAY TO REMEMBER

- Discrete label → Classification
- Continuous value → Regression

What is Overfitting and Underfitting?

Underfitting

The model is too simple to capture the patterns. It performs poorly on both training and test data.

Overfitting

The model learns the training data too well — including noise. Great on training data, poor on unseen test data.

REDUCE OVERFITTING BY

- Using more training data
- Simplifying the model
- Applying regularization
- Using cross-validation

Many ML models fail not due to algorithms — but due to overfitting.

QUESTION 04 · OF 15

Explain the Bias–Variance trade-off.

It explains why models make errors.

Bias

Error from overly simple assumptions. High-bias models fail to capture complex patterns and often **underfit**.

Variance

Error from being too sensitive to small changes in the data. High-variance models learn noise and tend to **overfit**.

A good model balances both. More complexity reduces bias but increases variance — and vice versa.

QUESTION 05 · OF 15

What is Cross-Validation and why is it used?

A technique to evaluate how well a model **generalizes** to unseen data. Instead of one train-test split, the data is divided into multiple subsets — train on some, test on the rest, repeat, then average the performance.

IT HELPS

- Detect overfitting
- Use data more efficiently
- Get a reliable estimate of performance

K-Fold cross-validation is the most commonly used method.

How do you evaluate a Classification model?

It takes more than just accuracy.

Accuracy

Overall correctness — but misleading for imbalanced data.

Precision

How many predicted positives were actually correct.

Recall

How many actual positives were correctly identified.

F1-score

Balances precision and recall.

In fraud detection, recall often matters more than accuracy — missing fraud is costly.

QUESTION 07 · OF 15

What is the ROC Curve and AUC?

ROC Curve

Shows how the true positive rate changes with the false positive rate at different threshold values.

AUC (Area Under the Curve)

Summarizes the ROC curve into a single number — how well the model distinguishes between classes.

READING AUC

- Close to 1 → strong model
- Close to 0.5 → random guessing

ROC-AUC is especially useful for imbalanced classification problems.

What is Feature Scaling and why does it matter?

It ensures numerical features are on a **similar scale**, so no feature dominates others due to its magnitude. Many algorithms rely on distance or gradient calculations.

Normalization

Scales values between 0 and 1.

Standardization

Centres data around mean 0 with unit variance.

Tree-based models usually do not require feature scaling.

Ashwini
Senior AI/ML Engineer

How does a Decision Tree make decisions?

It asks a series of simple questions about the data. At each **node** it picks one feature and a condition that best splits the data into purer groups (most points belong to the same class).

SPLIT METRICS

- Gini Impurity — how mixed the classes are
- Entropy — randomness in the data

PROS / CONS

✅ Easy to understand & explain

✅ Works with numerical & categorical

❌ Very prone to overfitting

❌ Unstable to small data changes

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What is a Random Forest, and why is it better than a Decision Tree?

An **ensemble** that combines many decision trees. Each is trained on random subsets of data and sees different features; the final output is majority voting (classification) or averaging (regression). Different mistakes cancel out.

WHY IT'S BETTER

- Reduces overfitting
- More stable & reliable
- Works well without heavy tuning

Trade-off: harder to interpret, slower and more memory-heavy.

What are L1 and L2 Regularization, and why use them?

Regularization controls model complexity and prevents overfitting by discouraging very large feature weights.

L1 — Lasso

Penalty = absolute value of weights. Can reduce some weights to zero → performs feature selection.

L2 — Ridge

Penalty = square of weights. Shrinks weights but rarely to zero → keeps all features, reduces their impact.

It prevents memorizing noise and encourages simpler, more general models.

What is Gradient Descent and how does it work?

An optimization algorithm that finds the best parameters by **minimizing a loss function** — like walking down a mountain: check the slope, step downward, repeat.

THE STEPS

- Start with random parameters
- Calculate the loss (error)
- Compute the gradient
- Move parameters the opposite way
- Repeat until loss stops decreasing

Learning rate: too high → may never converge; too low → very slow.

QUESTION 13 · OF 15

Why is the Train–Validation–Test split important?

It lets you **honestly evaluate** how well a model performs.

Training set

Used to teach the model patterns.

Validation set

Used to tune hyperparameters and make model decisions.

Test set

Used only once for final performance evaluation.

Proper splitting prevents data leakage and gives confidence the model works in the real world.

What is a Confusion Matrix and how do you read it?

A table of **actual vs predicted** values for a classification problem.

TP Correct positive	**FN** Predicted −, actually +
FP Predicted +, actually −	**TN** Correct negative

FROM IT WE DERIVE

- Accuracy
- Precision
- Recall
- F1-score

It reveals what kind of errors the model makes — far more useful than accuracy alone.

What is Feature Engineering and why is it important?

Creating **better input data** for a model. Raw data is rarely useful — feature engineering transforms it into a format models can learn from.

EXAMPLES

- Extracting day / month / weekday from a date
- Encoding categories like city or gender

WHY IT MATTERS

- Models learn from features, not raw data
- Good features drastically improve performance
- Often more impactful than changing algorithms

THE REAL EDGE

It's not about perfect math

Most interviewers don't expect advanced theory. They expect:

- ✅ Clear thinking
- ✅ Practical understanding
- ✅ Ability to explain simply

Explain these confidently and you'll outperform most 0–3 year candidates. Bookmark this — revise in minutes.

Ashwini
Senior AI/ML Engineer

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